

Numerical PDE Quiz 2

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Consider the 3D elliptic problem on a unit cube $\Omega = (0, 1)^3$ with a constant $\sigma > 0$:

$$\begin{cases} -\Delta u + \sigma u = f(x, y, z), & (x, y, z) \in \Omega, \\ u = 0, & (x, y, z) \in \partial\Omega. \end{cases}$$

Here $\Delta u = u_{xx} + u_{yy} + u_{zz}$. Let $h = \frac{1}{N}$ be the mesh size, and define the grid points $(x_i, y_j, z_k) = (ih, jh, kh)$ for $i, j, k = 0, 1, \dots, N$. Let $u_{i,j,k} \approx u(x_i, y_j, z_k)$ denote the numerical solution.

1. Write down the finite difference equations for the interior points $1 \leq i, j, k \leq N - 1$ using the standard 7-point stencil.
2. Rewrite the above finite difference equations into the matrix-vector form $AU = F$. Present the matrix A by using Kronecker product for short.
3. Write down eigenpairs of matrix A .
4. If $f(x, y, z) \leq 0$ for all $(x, y, z) \in \Omega$, show that $\max_{0 \leq i,j,k \leq N} u_{i,j,k} \leq 0$.
5. Define the space $H_0^1(\Omega) = \{u \in L^2(\Omega) : \nabla u \in L^2(\Omega), u|_{\partial\Omega} = 0\}$. Write down the weak formulation: find $u \in H_0^1(\Omega)$ such that $a(u, v) = L(v)$ holds for any $v \in H_0^1(\Omega)$. Explicitly write down the bilinear form $a(\cdot, \cdot)$ and the linear functional $L(\cdot)$.